

?? with a thickness d and refractive index N_2 which acts like a fabry-perot interferometer. Light is sim-

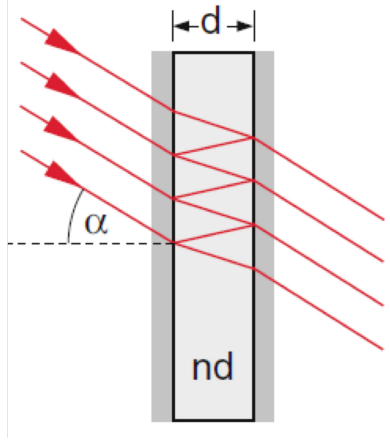


Abbildung 1: Simplified illustration of the measured samples as a plane parallel plate with thickness d and refractive index N_2 . The surrounding air is described with a refractive index $N_1 \approx 1$ [?]

plified as beams. Every incoming beam is partially reflected and partially transmitted at the first interface. The total power reflectivity R_1 is described with:

$$R_1 = \left| \frac{1 - N_2}{1 + N_2} \right|^2 \quad (4)$$

without considering the phase shift of the reflected beam. Like shown in the illustration, the total power reflectivity of the second interface R_2 consists of the incoming beams and the beams reflected at the first interface within the material. Additionally, the absorption of the material is considered. Therefore, R_2 is described with:

$$R_2 = R_1(1 - R_1)^2 \cdot e^{-2\beta d} \quad (5)$$

with the absorption coefficient β and the total power reflectivity R_1 of the first interface. The assumption that the reflection of the first interface is the same as the reflection of the second interface is in reality not correct. Disturbances like roughness of the interfaces or the absorption of the material lead to a different reflection coefficients of the two interfaces.

Considering this and the phase of the light, the total power reflectivity R is described with:

$$R = \left| \frac{(e^{i\frac{2\omega}{c}N_2d} - 1)(1 - N_2)}{e^{i\frac{2\omega}{c}N_2d}(1 - N_2) - (1 + N_2)} \right|^2 \quad (6)$$

ω is the angular frequency of the light and c the speed of light. The phase shift of the light is described with $e^{i\frac{2\omega}{c}N_2d}$. In contrast to ??, equation ?? shows a periodic modulation of the reflectivity because of interference. The difference between the maxima of the

modulation can be calculated with:

$$\Delta\nu = \nu_{m+1} - \nu_m = \frac{1}{2dN_2} \quad (7)$$

The refractive index N_2 can be calculated either with equation ?? or with

$$N_2 = \sqrt{\epsilon} \quad (8)$$

using the square root of the dielectric function ϵ of the material. The dielectric function ϵ is described in the drude model as:

$$\epsilon = 1 - \frac{\omega_p^2\tau^2}{1 + \omega^2\tau^2} + i\frac{\omega_p^2\tau}{\omega(1 + \omega^2\tau^2)} \quad (9)$$

with ω_p as the plasma frequency:

$$\omega_p = \sqrt{\frac{N_e e^2}{\epsilon_0 m^*}} \quad (10)$$

and τ as the relaxation time of the electrons. m^* is the effective mass of the electrons and N_e the charge carrier density.

For semiconductors, ϵ_{sem} can be calculated as:

$$\epsilon_{\text{sem}} = \epsilon_{\text{LP}} + i\frac{\sigma}{\omega\epsilon_0} \quad (11)$$

with σ as the conductivity of the semiconductor:

$$\sigma = \frac{N_e e^2 \tau}{m^*} \frac{1}{1 - i\omega\tau}. \quad (12)$$

The lattice contribution ϵ_{LP} is described as:

$$\epsilon_{\text{LP}} = \epsilon_{\infty} \left(1 + \frac{\omega_{\text{LO}}^2 - \omega_{\text{TO}}^2}{\omega_{\text{TO}}^2 - \omega^2 - i\Gamma\omega} \right) \quad (13)$$

with ϵ_{∞} as the high frequency dielectric constant, ω_{LO} as the longitudinal optical phonon frequency, ω_{TO} as the transversal optical phonon frequency and Γ as the damping constant.

3 Set-Up

Fig. ?? shows a simplified schematic version of the *Bruker IFS 66 v/s* fourier spectrometer. The source (1) emits light in the near-infrared range ($1500 \text{ cm}^{-1} - 15000 \text{ cm}^{-1}$) which is then reflected on the beam splitter (3). The Beamsplitter, static mirror (4) and movable mirror (5) can be seen as a michelson interferometer which is used to create an interference pattern. During the measurement, the movable mirror is moved with constant speed. The light is then focused via another mirror (6) on the sample (7). In this case, transmission is shown. For reflection measurements an additional module is used which is not shown in the figure. Mirror (8) then focuses the light onto the detector (9). The signal is then digitalized. Numbers (10) to (13) show the beam path of a laser which is used to determine the position of the movable mirror.

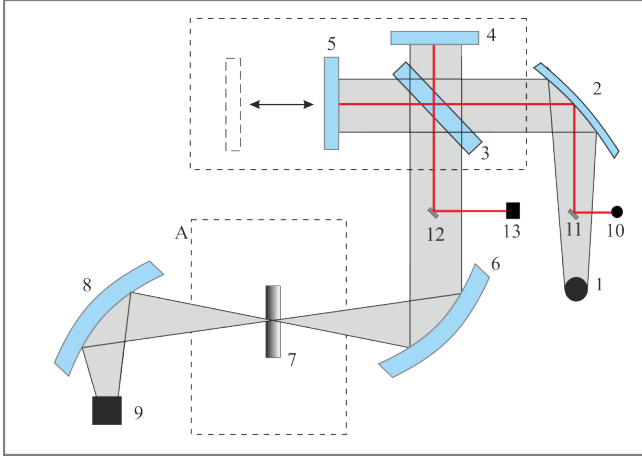


Abbildung 2: Simplified schematic illustration of the set-up used for all experiments. A short overview of the working principle is given in the text.[?]

4 Experiment

All following transmission and reflection experiments are done at ambient air in the chamber. The evaluation of the signal with a 'Mertz' phase correction is done automatically by the system. Furthermore, the spectral range for all measurements is set to: $(1500 - 15000) \text{ cm}^{-1}$.

4.1 Gas absorption

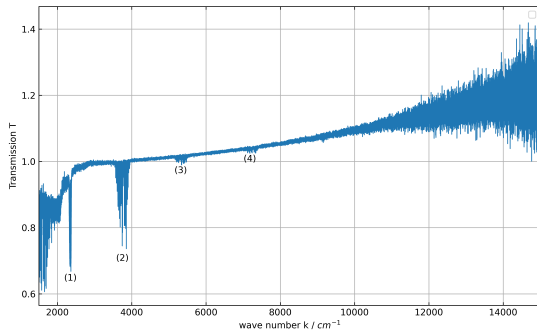


Abbildung 3: Normalized transmission of an empty chamber filled with ambient air. The spectral ranges for gas absorption are highlighted.

To identify the spectral ranges where the result is influenced by the gas absorption, a measurement at ambient air with a resolution of 0.3 cm^{-1} and $N = 10$ scans is performed. The measurement at air is normalized. The corresponding reference measurement is done under vacuum. One expects gas absorption dips at frequencies that are similar to the resonance frequencies of the air molecules. The wavenumbers of

the measured gas absorption is shown in tab ?? All

CO ₂	H ₂ O CO ₂	H ₂ O	H ₂ O
2350 cm ⁻¹	3750 cm ⁻¹	5320 cm ⁻¹	7180 cm ⁻¹

Tabelle 1: Regions of strong gas absorption in fig ??.
The measured values agree with literature values for H₂O and CO₂ [?].

the measured data agrees with the literature values for the gases H₂O and CO₂.

4.2 Signal to noise ratio

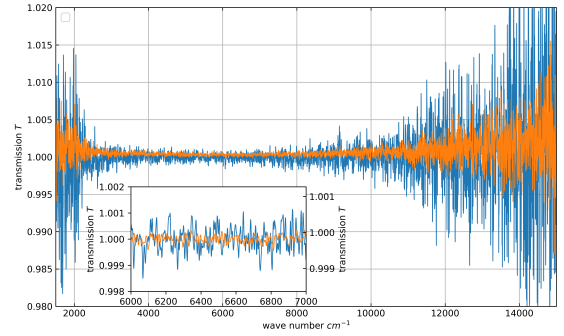


Abbildung 4: Transmission measurements for $N_{\text{scan}} = 10$ and $N_{\text{scan}} = 100$ mirror passes. The inset shows the part used for evaluation of the signal to noise ratio (see fig. ??)

To determine the influence of the number of scans N_{scan} on the signal to noise ratio, measurements with a resolution of 4 cm^{-1} are done at various N_{scan} . Those are transmission measurements with ambient air. Between 6000 cm^{-1} and 7000 cm^{-1} a region with generally less noise (see fig. ??) and no gas absorption (see fig. ??) is identified. To quantitatively examine the behaviour of the signal to noise ratio, the mean value T_{mean} is divided by the Standard deviation T_{std} for every transmission measurement. Thus the signal to noise ratio SNR is given by:

$$SNR = \frac{T_{\text{mean}}}{T_{\text{std}}} \quad (14)$$

Theoretically, one expects SNR to grow according to:

$$SNR = N^a \quad (15)$$

with $a = 0.5$ [?]. To check this dependency, the signal to noise ratio is plotted for various N 's in a double logarithmic scale. From the slope of the linear regression a can be determined as:

$$a = 4.967 \cdot 10^{-1} \pm 0.025 \quad (16)$$

which matches with the expectation.

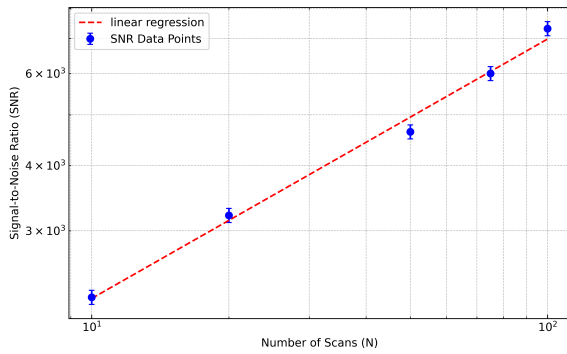


Abbildung 5: SNR over the number of mirror passes in double logarithmic scaling using the spectral range between 6000 cm^{-1} and 7000 cm^{-1}

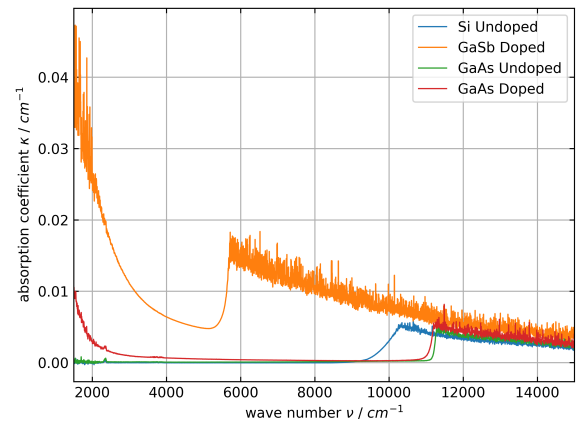


Abbildung 7: Extinction coefficient κ of different semiconductor samples.

4.3 Transmission at high resolution

4.4 Reflection and transmission at low resolution

4.5 Refractive index N

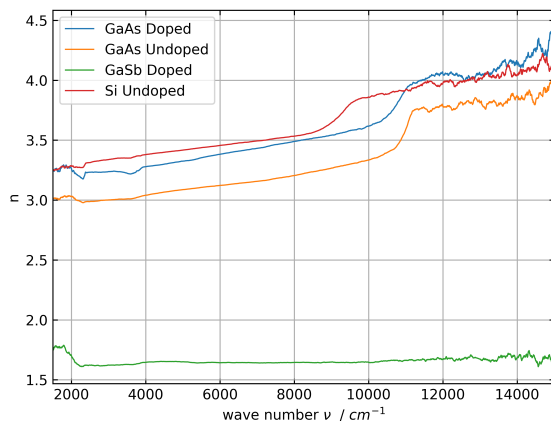


Abbildung 6: Refractive index n of different semiconductor samples.

4.6 Bandgap and pulse matrix element

5 Bibliography

Literatur

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[5] ?????????????????

[6] ????????

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Images/betas.png

Abbildung 8: Absorption coefficient β of different semiconductor samples.

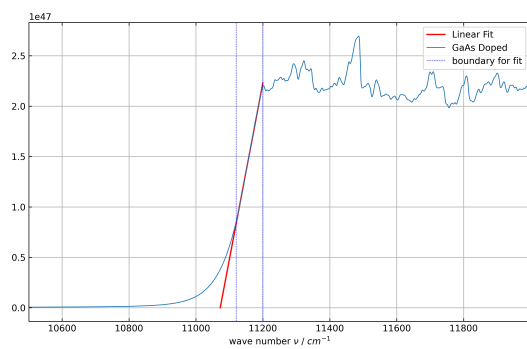


Abbildung 9: Determination of the bandgap.